Nishant Kumar

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SUMMARY

Final-year B.Tech Computer Science student with experience in backend engineering, distributed processing, and data-intensive applications. At **Comviva Technologies**, developed production-grade systems for music catalog comparison, job orchestration, caching, and parallel processing using Python, Redis, MySQL, SQLAlchemy, and REST APIs. Proficient in React, Node.js, and database design. Passionate about building scalable backend systems and seeking Software Development Engineer roles.

EDUCATION

Raj Kumar Goel Institute Of Technology **2022 – 2026**
B.Tech - Computer Science and Engineering (Data Science) - ***CGPA - 7.3*** *Ghaziabad, Uttar Pradesh*

EXPERIENCE

Software Development Intern | Comviva Technologies Ltd. **Feb 2026 – Present**
CQI Processor | Python, Redis, MySQL, RapidFuzz, Loguru *Gurugram, India*

- Engineered a multi-platform **CQI Processor** matching catalog songs across **4 streaming platforms** (Apple Music, Spotify, YouTube, Shazam) using RapidFuzz fuzzy matching and ISRC-based exact lookup, achieving **95% match accuracy** on large music catalogs.
- Designed a configurable **scoring engine** computing **3-dimensional CQI scores** (Availability, Correctness, Accuracy) with per-platform weight tuning, enabling data-driven quality benchmarking across catalog entries.
- Implemented a **Redis cache-first strategy** with dynamic midnight-expiry TTL, reducing redundant platform API calls by ∼**60%** across concurrent jobs and cutting average job latency on repeat catalog runs.
- Built a **MySQL job scheduler** managing a **4-state lifecycle** (`CREATED` → `IN_PROGRESS` → `COMPLETED/FAILED`) with SQLAlchemy connection pooling, supporting reliable concurrent job execution without race conditions.
- Engineered a two-tier **parallel pipeline** using `ProcessPoolExecutor` (per song) and `ThreadPoolExecutor` (per platform), reducing end-to-end catalog comparison time by ∼**50%** vs. sequential execution on **10,000+ song** workloads.
- Integrated **Apple Music (JWT auth)** and **SoundCharts REST API** clients with per-region chart URL resolution; generated structured per-platform **CSV reports** with field-level quality scores consumed by downstream analytics pipelines.

PROJECTS

Full-Stack Food Delivery Platform **Nov 2024 – Mar 2025**
React, Vite, React Router, MongoDB

- Architected a full-stack **MERN** application using React, Vite, and MongoDB to handle end-to-end user workflows including authentication, food catalog, and order management.
- Engineered a secure **User Management** system with encrypted registration, login, and profile editing using React hooks for real-time state management.
- Developed advanced **Search and Filter** algorithms to categorize items by price and attribute, with a persistent Order Cart ensuring **100% accuracy** in item quantities.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), SQL
Web Technologies: ReactJS, Next.js, Node.js, HTML5, CSS3, REST APIs
Databases & Backend: MongoDB, MySQL, Redis, SQLAlchemy
Tools & DevOps: Git, GitHub, Postman, Jenkins, Linux, VS Code
CourseWork: Data Structures & Algorithms, Database Management System

CERTIFICATE

- Web Development Intern – Codveda
- Data Engineering and Visualization – HCLTech
- Course Participation on Data Analytics – AWS
- Workshop on Frontend Development – ScholarHat

EXTRACURRICULAR

- Actively solve DSA problems on **LeetCode** and **GFG** (150+ problems), strengthening problem-solving and algorithmic thinking skills.
- Participated in the **Inter-College Debate Competition**, enhancing public speaking and critical thinking abilities.